CLIP to Grounding DINO

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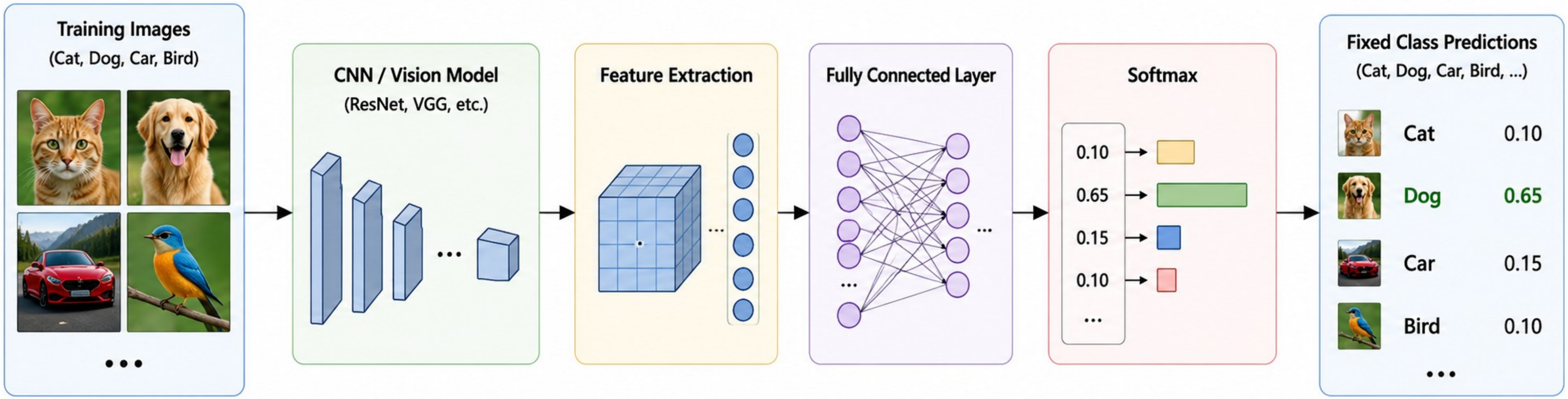
Traditional image classifiers

In **traditional image classification**, the model learns a mapping from images to a **fixed set of predefined categories**. During training, each image is associated with a class label, and the network learns discriminative visual features for those classes only.

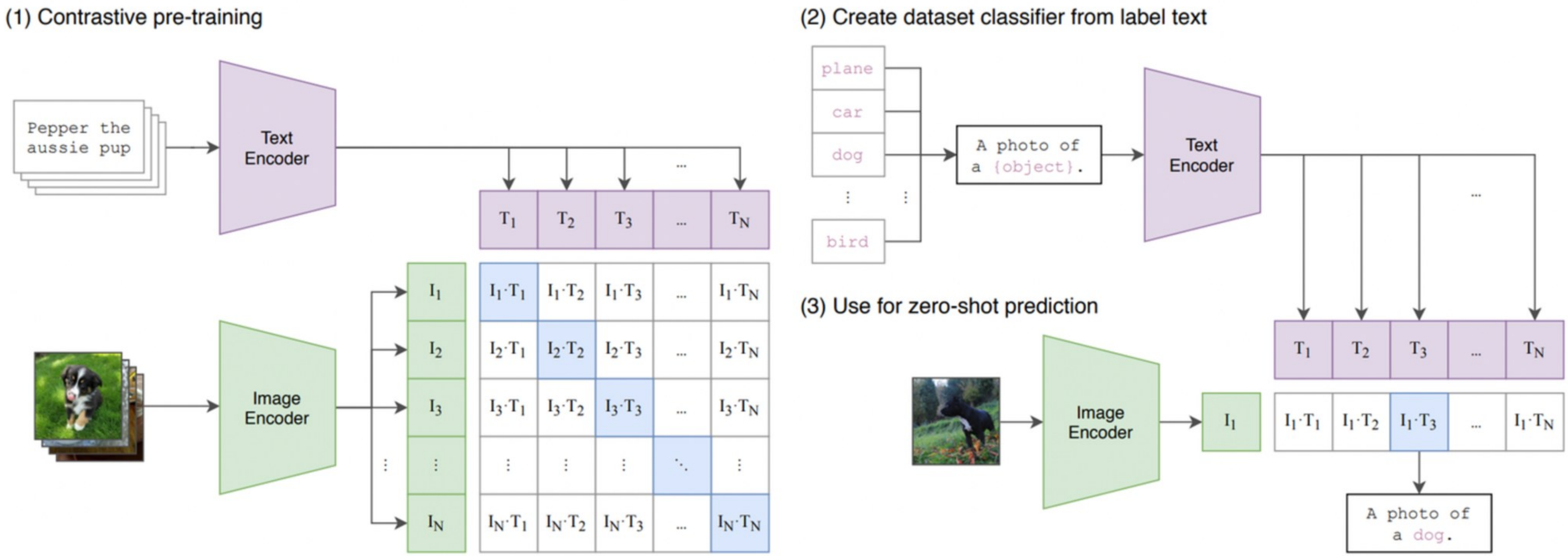


Limitations of Traditional Image Classification

1. Fixed Categories (Closed-Set Problem): Traditional classifiers can recognize only the classes seen during training.
2. Requires Large Labeled Datasets: Training deep networks needs thousands of labeled examples
3. Poor Zero-Shot Capability: Traditional models cannot classify new categories without retraining.
4. Weak Semantic Understanding: Traditional classifiers learn (Image → Class Label) rather than Image ↔ Language Meaning
5. Domain Shift Problem: Performance drops when the test distribution differs from training data.
6. Classifier Layer Depends on Number of Classes
7. No Text-Image Alignment: A traditional classifier cannot answer **Find a Red Car?.**
8. Cannot Perform Grounding or Segmentation: Where the dog is?, Bounding box coordinates, and Segmentation

What is CLIP?

CLIP = Contrastive Language Image Pre-training



What is CLIP?

CLIP a neural network that jointly trains an image encoder and text encoder to map respective modalities to the same embedding space

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Let's understand how and why we do this...

Training CLIP

Consider a training set of image + free text pairs

This mural of a siberian huskey is so beautiful

The 2012 Honda accord has 160K miles on it.

My cute sally is the best dog in the world.

We got lucky seeing antelope fighting.

Training CLIP

Create batches of size N.

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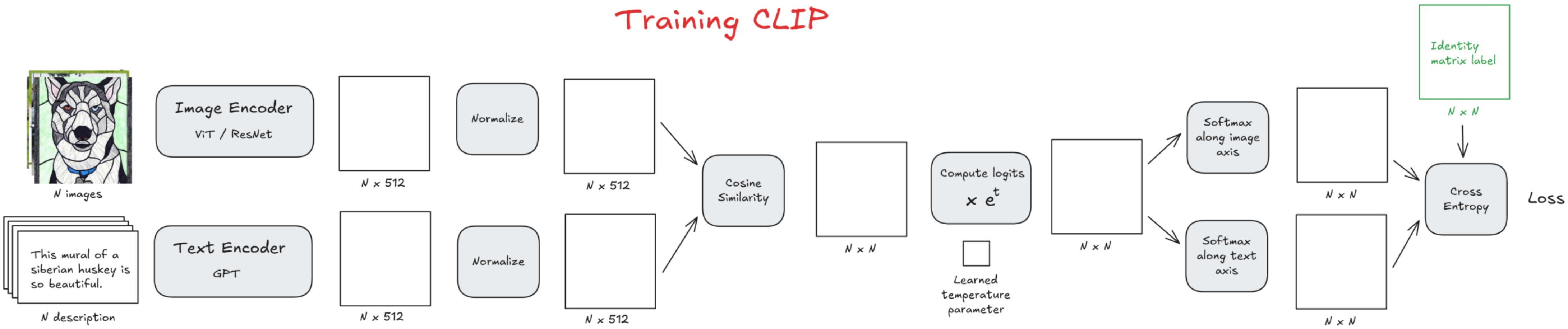
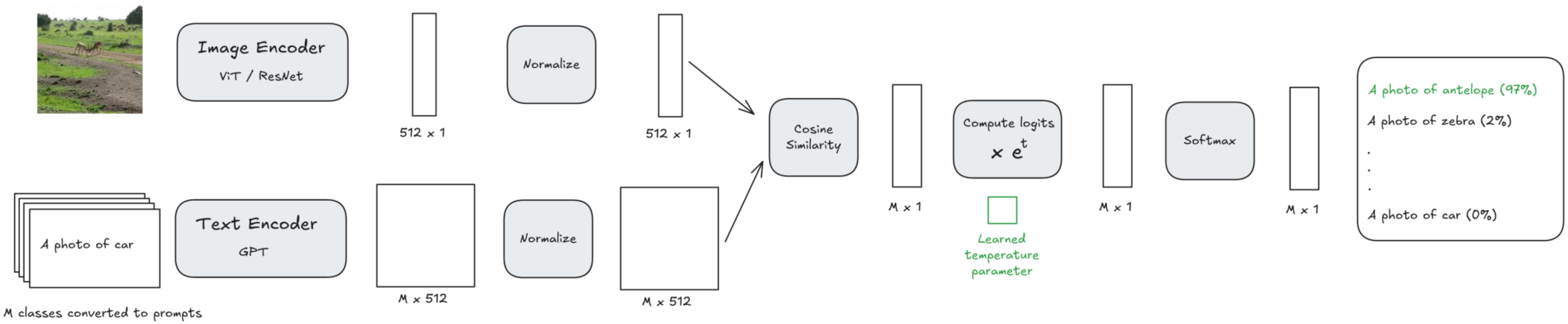
Training CLIP



Zero Shot Inference



Why CLIP?

1. Task specific labels can be hard to come by

Dog

Car

Dog

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1. Task specific labels can be hard to come by

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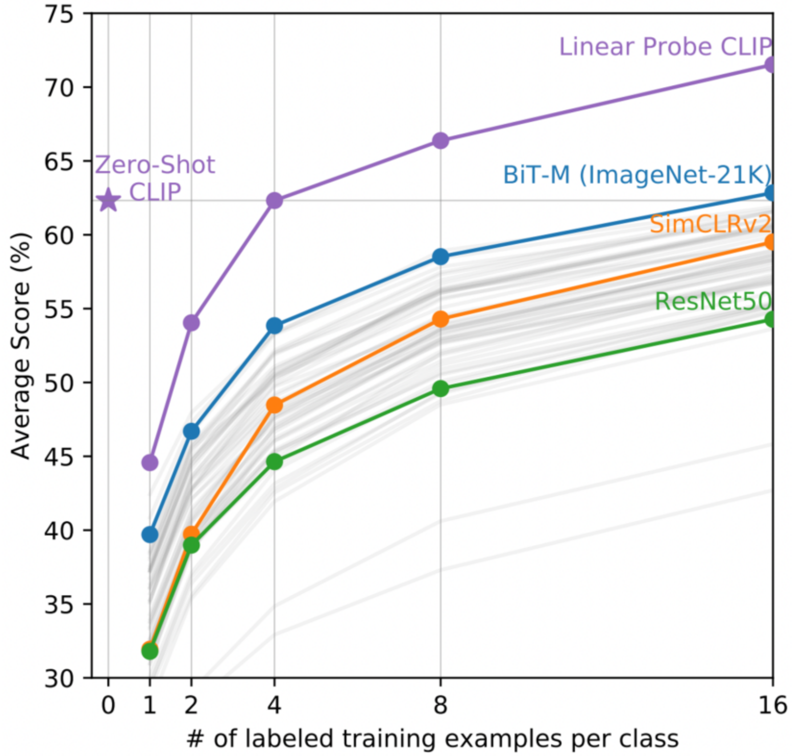
It is far easier to get natural language labels

Why CLIP?

2. Natural Language supervision allows the image encoder to create rich vectors that better encode the meaning of the image.

Let's take a look at some code to show how

Performance



During inference, Zero shot CLIP outperforms other networks (convolutional or transformer) that are trained with golden labels.

CLIP_Tutorial_Scratch.ipynb

STTP_CLIP.ipynb

https://github.com/praveenkumarchandaliya/STTP_Program/tree/MetaLearning

Linear Probing

Method to evaluate the visual representations of CLIP Image encoder. It involves training a linear model (probe) on top of the frozen CLIP encoder.

Linear Probing

Say we have a dataset with classification labels

Dog

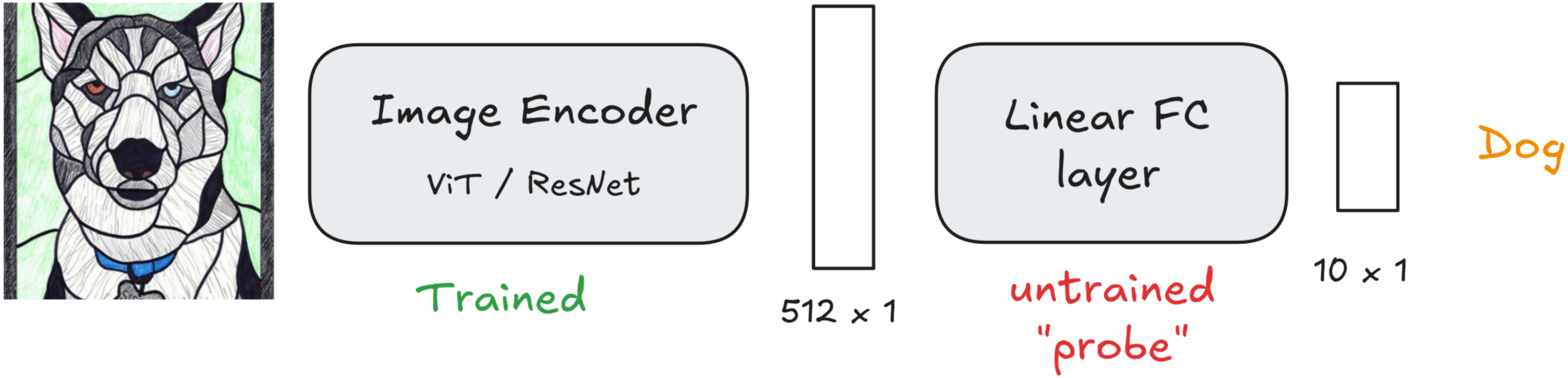
Car

Dog

Say we have a 10 class classification task

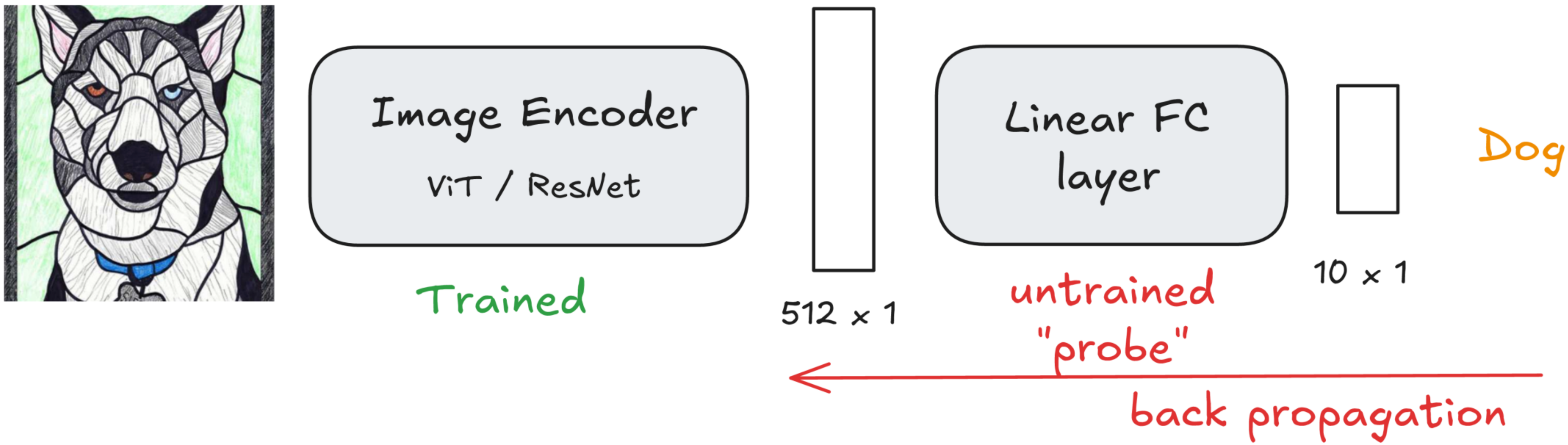
Linear Probing

Add a linear feed forward layer and train



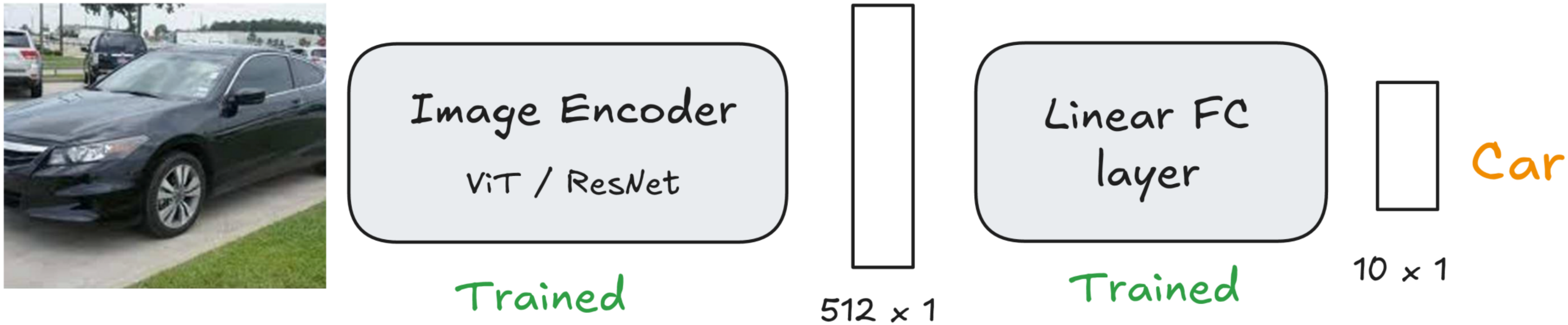
Linear Probing

Freeze CLIP encoder weights and update only the FC weights



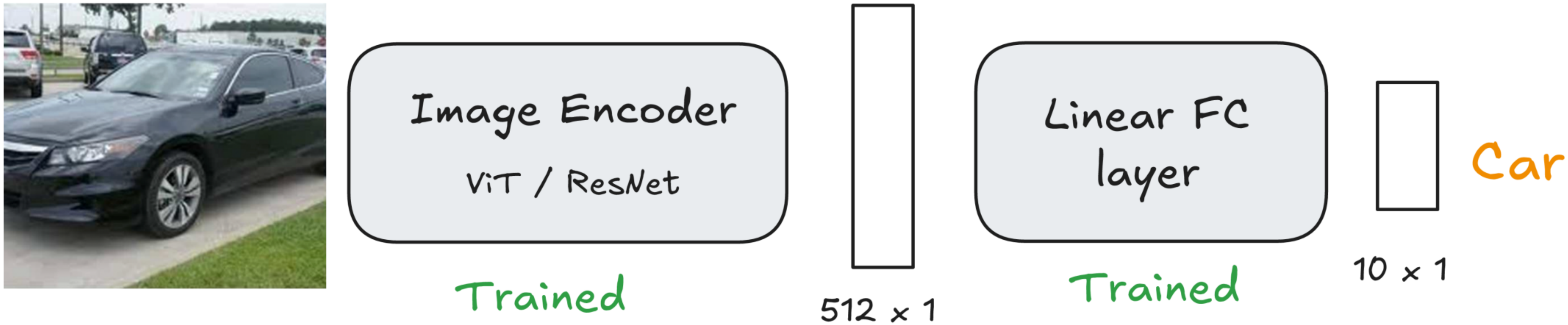
Linear Probing

Once trained, the probe can help understand how well the CLIP encoder performs on the new dataset.



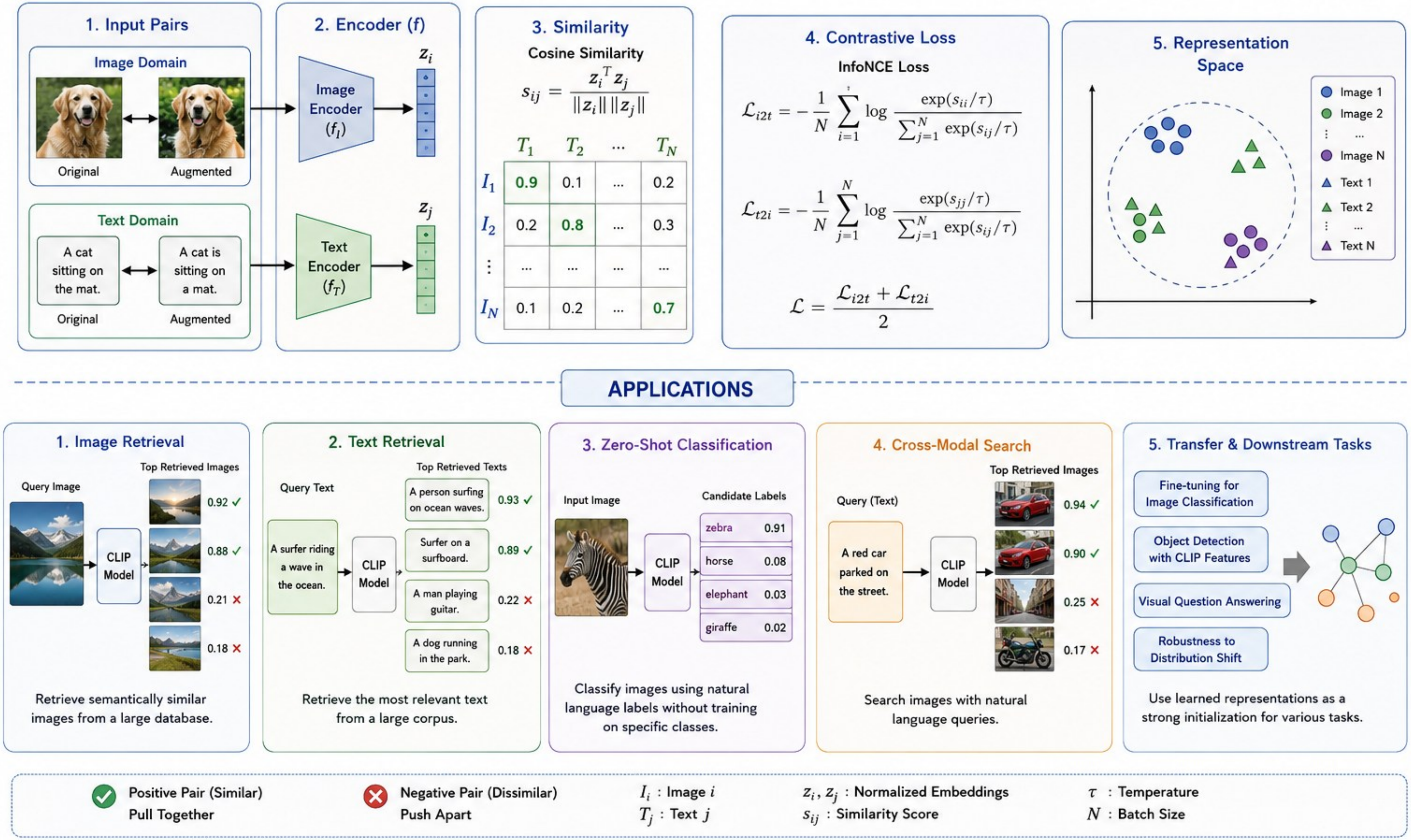
Linear Probing

It can even improve performance of CLIP Encoder on that specific dataset if data for each category is easy to come by.



CONTRASTIVE LEARNING: APPLICATIONS

Contrastive learning learns representations by pulling similar pairs together and pushing dissimilar pairs apart.



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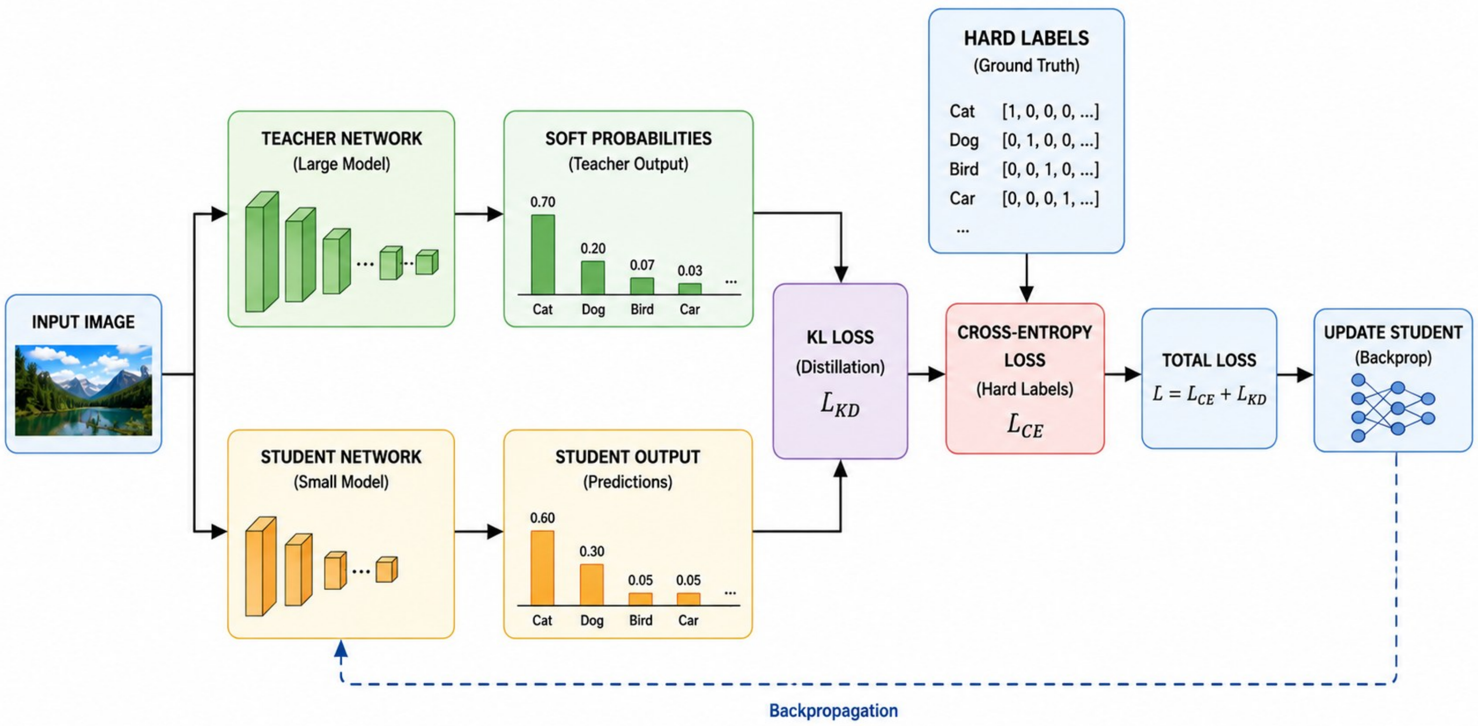
What is Knowledge Distillation?

Knowledge Distillation (KD) is a model compression and transfer learning technique in which a large, well-trained **Teacher Network** transfers its knowledge to a smaller **Student Network**.

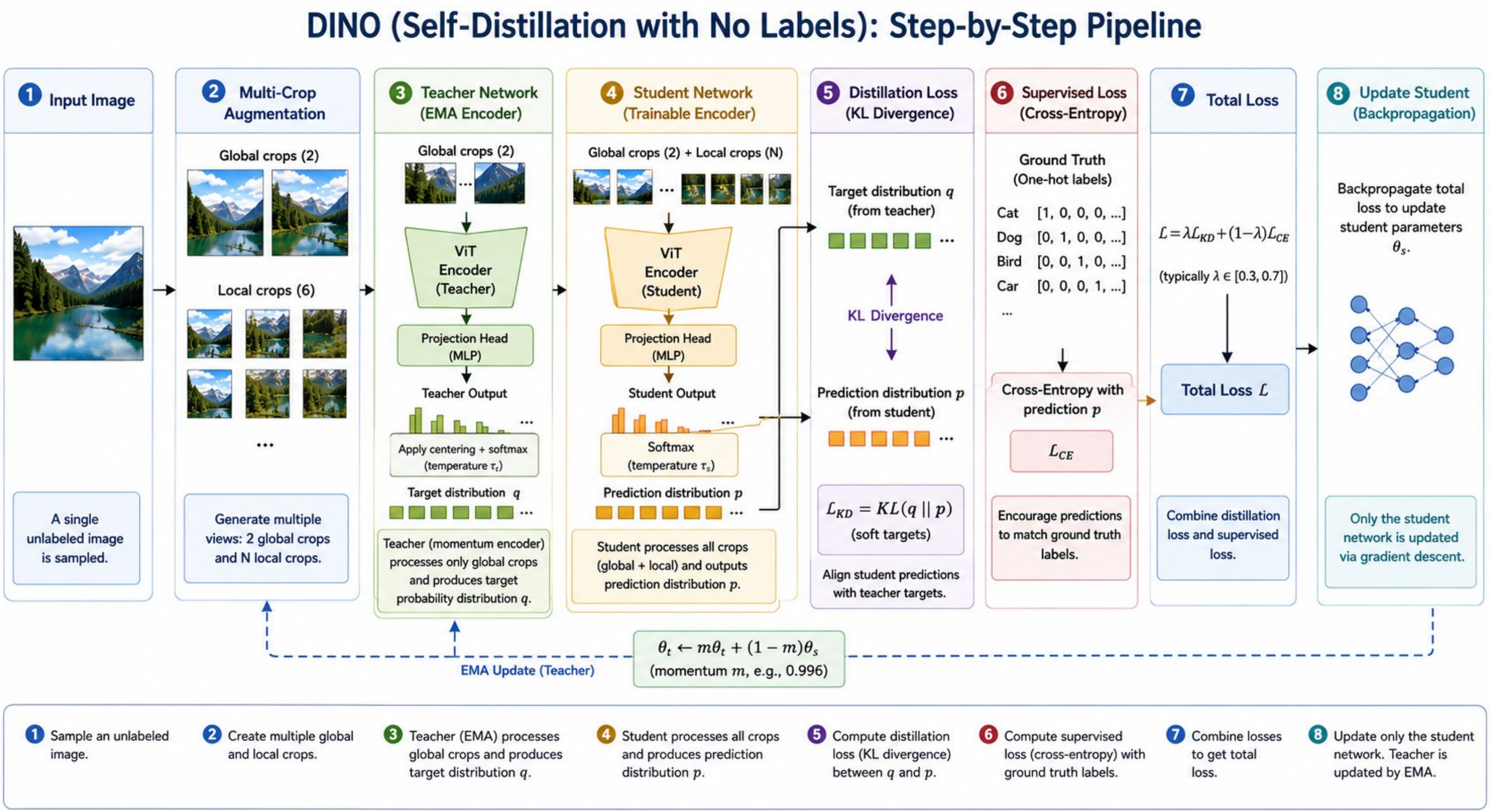
The **primary architecture of Knowledge Distillation** is the **Teacher–Student framework**, where:

1. A large **Teacher Network** is trained.
2. The teacher generates **soft targets**.
3. A smaller **Student Network** learns from both:
    1. ground-truth labels, and
    2. teacher predictions.
4. The student achieves performance close to the teacher with fewer parameters and lower computational cost.



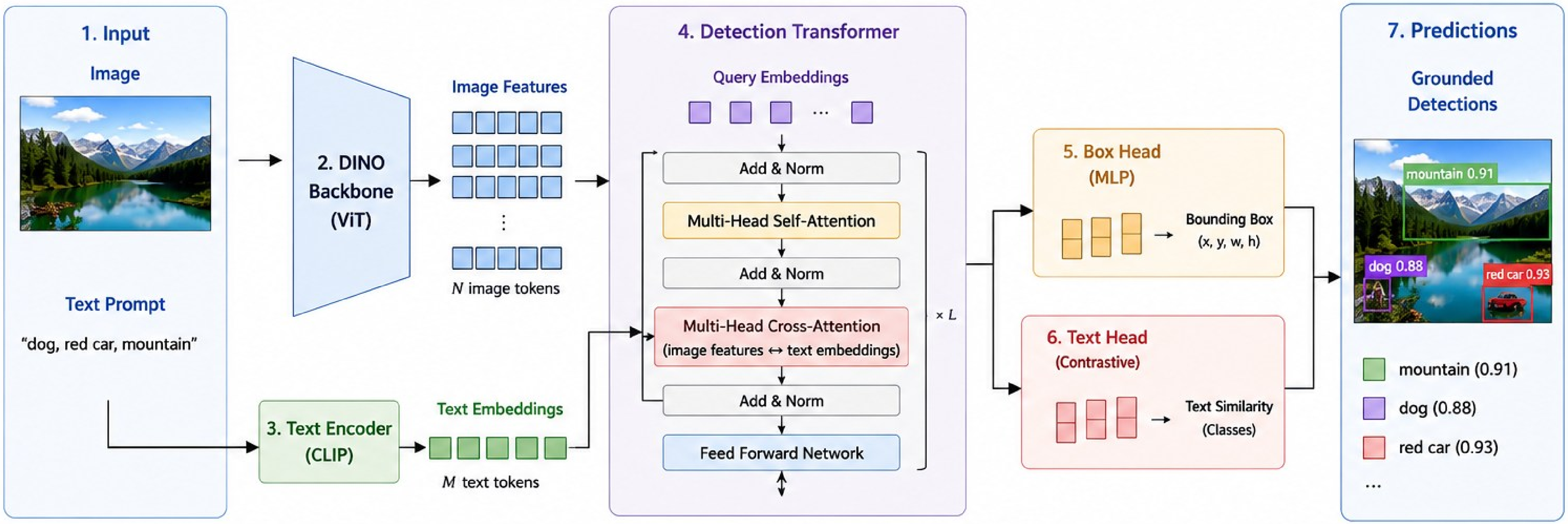
DINO (Self-Distillation with No Labels)

DINO learns powerful visual representations **without labels** by training a **Student Network** to mimic a **Teacher Network**. Only the student is updated by backpropagation, while the teacher is updated using an **Exponential Moving Average (EMA)**.



Ground DINO Architecture

Open-vocabulary object detection with grounded text



1. Input

- An image and a text prompt containing object names.

Text: "dog, red car, mountain"

2. DINO Backbone (ViT)

- Pretrained DINO extracts dense visual features from the image.
- Output: N image tokens.

3. Text Encoder (CLIP)

- CLIP text encoder converts text prompt into embeddings.
- Output: M text tokens.

4. Detection Transformer

- Query embeddings interact with image features and text embeddings.
- Self-attention models query relations.
- Cross-attention aligns queries with both image and text.
- Repeated for L layers.

5. Box Head

- MLP predicts bounding box (x, y, w, h) for each query.

6. Text Head

- Predicts similarity between each query and text embeddings (contrastive).
- Enables open-vocabulary classification.

7. Predictions

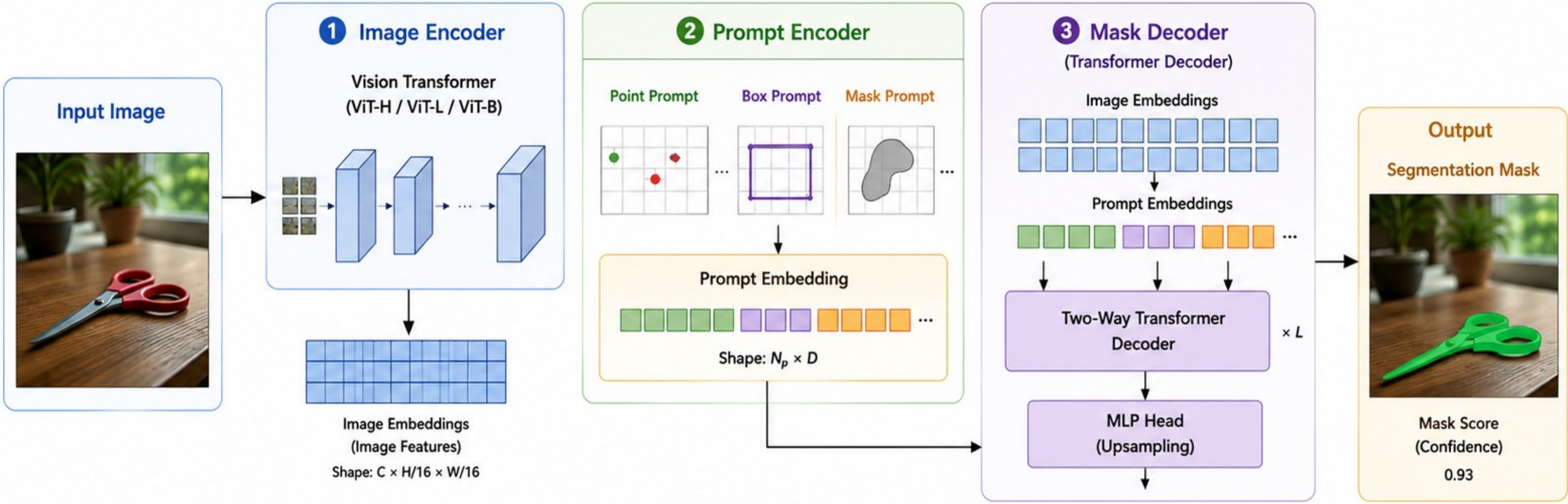
- Combine box and text scores.
- Keep high-confidence detections.
- Output grounded object boxes with labels.

Key Characteristics

- Open-vocabulary: no need to train on fixed categories.
- Grounded detection: links objects to text via contrastive alignment.
- Leverages DINO's strong self-supervised visual representations.
- Generalizes to unseen objects described by text prompts.
- Trained on image–text pairs with box annotations.

Segment Anything Model (SAM) Architecture

SAM architecture consists of three main components



Three Main Components

1 Image Encoder

- A pretrained Vision Transformer (ViT) extracts dense image features.
- Converts the input image into a grid of image embeddings.
- Output shape: C × H/16 × W/16

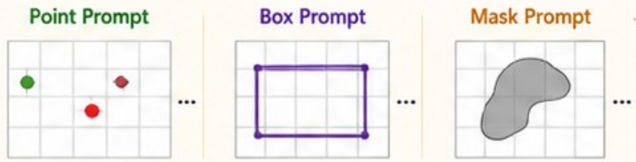
2 Prompt Encoder

- Encodes user prompts (points, boxes, masks) into prompt embeddings.
- Different prompt types are embedded into a unified representation.
- Output shape: $N_p \times D$

3 Mask Decoder

- A transformer decoder that performs two-way attention between image embeddings and prompt embeddings.
- Predicts the segmentation mask for the object(s) specified by the prompt(s).
- Output: high-quality segmentation mask + score

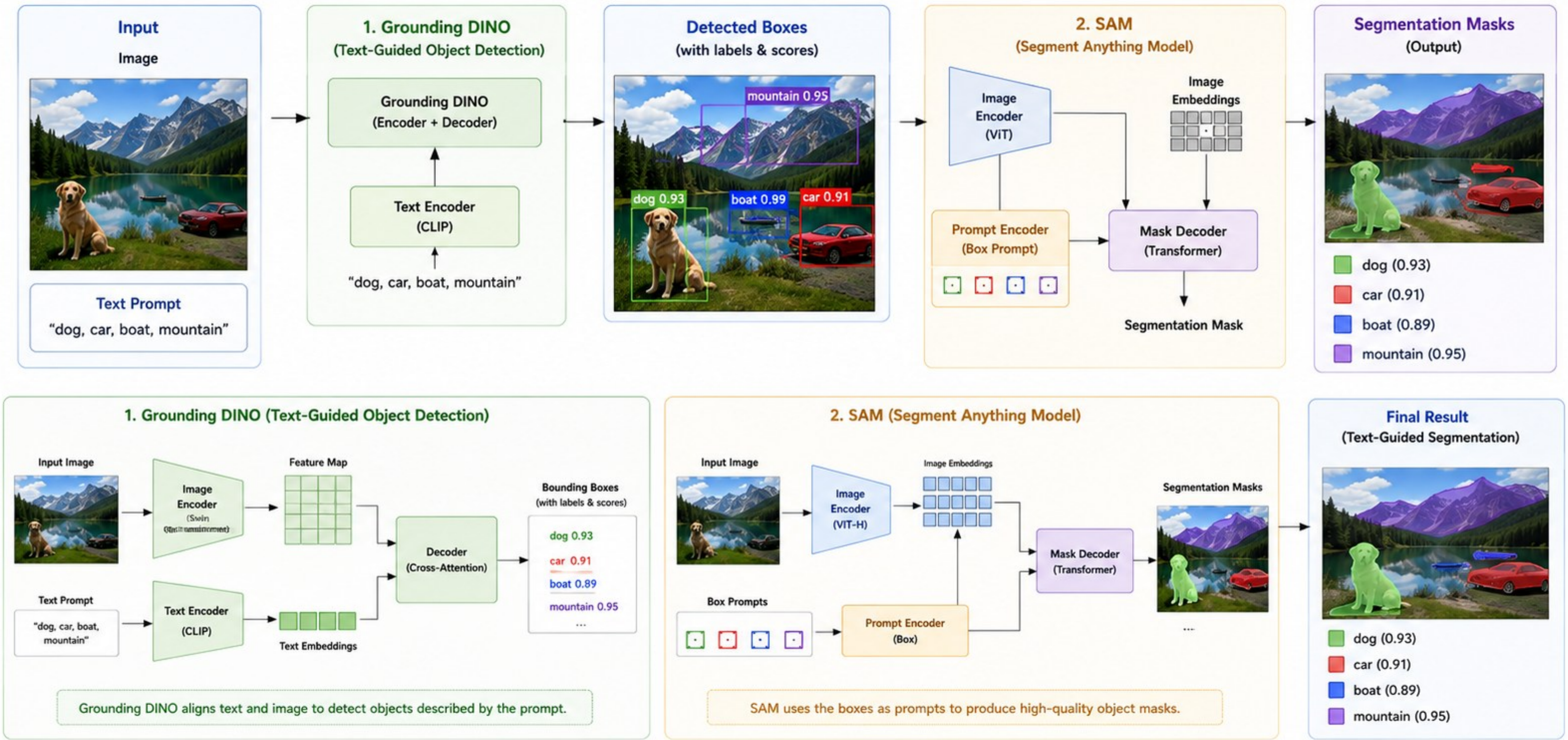
Supported Prompts



You can use multiple prompts together.

SAM + Grounding DINO: Text-Guided Segmentation

Use Grounding DINO to detect objects from text, then SAM to segment them precisely.



Why Combine SAM + Grounding DINO?

- ✓ Grounding DINO localizes objects from text (open-vocabulary detection).
- ✓ SAM produces pixel-accurate masks for the detected objects.
- ✓ Works for any text prompt without training on specific categories.
- ✓ Powerful for open-world segmentation, image editing, robotics, and more.

Grounding DINO

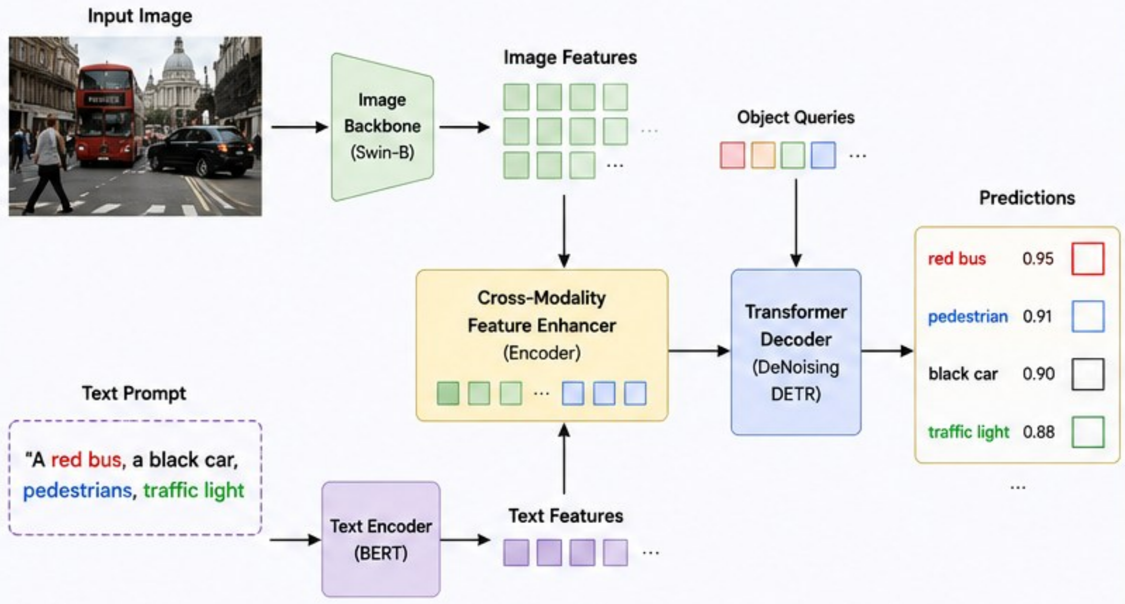
Open-Set Object Detection with Language Guidance

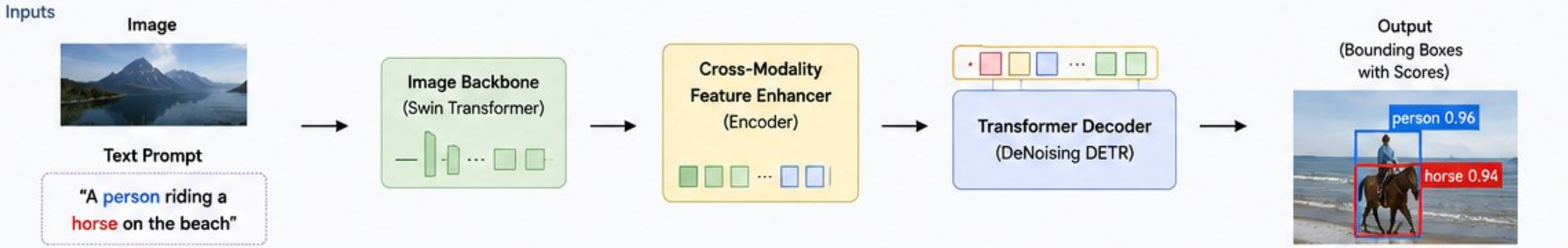
Grounding DINO is a vision-language foundation model that can detect arbitrary objects in images using text prompts.

Highlights

- ✓ Open-set / open-vocabulary detection
- ✓ Text-prompt driven
- ✓ No need for task-specific training
- ✓ Strong zero-shot transferability

Architecture



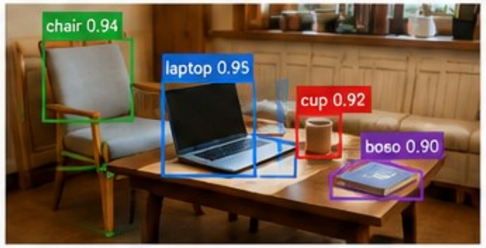


Detection Results (Zero-Shot)

Prompt: "A red bus, a black car, pedestrians, traffic light"

Prompt: "dog, skateboard"

Prompt: "chair, laptop, cup, book"



Key Components

Image Backbone
Extract visual features from the image.

Text Encoder
Encode the text prompt into language features.

Cross-Modality Feature Enhancer
Fuses image and text features via multi-head cross-attention.

DeNoising DETR Decoder
Uses learnable object queries and iterative refinement to predict boxes.

Language-Guided Query Selection
Object queries are initialized and guided by the text semantics.

Matching & Loss
Bipartite matching with L1 + GIoU + Focal loss and contrastive grounding loss.

Training Data

Large-scale image-text paired data (Object365, GoldG, Cap4M, etc.)

Grounding annotations for boxes and phrase alignments.

Trained in a single stage end-to-end.

Key Features

- ✓ Open-vocabulary: detect any text-described object
- ✓ Zero-shot generalization to unseen categories
- ✓ No task-specific fine-tuning required
- ✓ Works with complex language expressions
- ✓ Can be combined with SAM for pixel-level masks (Grounded-SAM)
- ✓ Strong performance on many benchmarks (ODinW, GRIT, RefCOCO, etc.)

Applications

- Robotics & Vision-Language Agents
- Autonomous Driving
- Remote Sensing
- Medical Imaging
- Human-Computer Interaction
- Image/Video Understanding

Model	Task	Input
CLIP	Image-text alignment	Image + Text
SAM	Segmentation	Image + Prompt
Grounding DINO	Object detection	Image + Text
BLIP	Captioning/VQA	Image + Text
Florence-2	Unified vision tasks	Image + Text
OWL-ViT	Open-vocabulary detection	Image + Text